

National Institute of Technology Kurukshetra
Knowledge Representation and Problem Searching-AIPC 201(AIML)
Mid-Term Examination-II, 3rd Semester, 2025-26

Time: 50 Minutes

Total Marks: 15

Note: Attempt all the questions. If require any missing data; then choose suitably.

Q1. Assume predicate: $\text{Parent}(x, y)$ which means x is parent of y , $\text{Female}(x)$ means x is female. David and Allena are two constants. $\exists^1$ is used to represent, there exist exactly one. Express the following in first order logic.

- (i) David has exactly one child, daughter.
- (ii) David and Allena have exactly one child together.
- (iii) David has at least one child with Allena, no child with others.
- (iv) For some child of David, all flowers are beautiful.

[4 marks]

Q2. Consider the following sentences.

S1: Everyone who loves all animals is loved by someone, S2: Anyone who kills an animal is loved by no one, S3: Jack loves all animals, S4: Either Jack or Jim killed a cat, named Tuna, S5: All the cats are animals. Convert these sentences to first order logic sentences.

Given are the predicates; $\text{Loves}(X, Y)$: X loves Y , $\text{Animal}(X)$: X is an animal, $\text{Kill}(X, Y)$: X killed Y , $\text{Cat}(X)$: X is a cat. Use inference to verify if Jim killed Tuna by detailing each step and mentioning which inference rule is applied in each step.

[5 marks]

Q3. Given the following data containing x and $f(x)$:

x	-2.3	-1.2	-0.3	1.4	2.6	3.9
$f(x)$	13.5	5.8	3.6	1.2	2.8	10.7

Use polynomial regression model of order 2 ($q_0 + q_1x + q_2x^2$) and calculate the best value of the parameters q_0, q_1, q_2 using batch gradient descent with learning rate $= 0.02$. Take initial value of $q_0 = 1.5, q_1 = -2.5, q_2 = 0.5$. Continue till two iterations. Take all the data as training data. After completion of the update of parameters after two iterations, find the MAPE value with respect to the given data.

[6 marks]